

Muhammad Saad Hamid

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PROFESSIONAL SUMMARY

Electrical Engineering undergraduate at FAST NUCES (Computer Engineering track) with hands-on expertise in embedded systems, digital logic design, and C/C++ programming. Demonstrated ability to design, simulate, and deploy hardware-software solutions — from biometric security systems to automated control circuits. Seeking a challenging engineering internship or entry-level role to deliver impactful, production-ready work.

EDUCATION

B.Sc. Electrical Engineering (Computer Engineering Track)

Expected June 2027

FAST NUCES — Lahore, Pakistan

Relevant Coursework: Digital Logic Design · Circuit Analysis · Programming Fundamentals · OOP · Linear Algebra · Feedback Control Systems · Electro-Mechanical Systems · Engineering Economics

TECHNICAL SKILLS

Languages: C, C++

Embedded Systems: PIC16F877A, Microcontrollers, VHDL (FSM design, Xilinx ISE), Digital Logic Design

Simulation & EDA: Proteus, MATLAB, Simulink, PSpice, Xilinx ISE

Dev Tools: Visual Studio, Linux (VMware), AutoCAD (Basic)

PROJECTS

Fingerprint-Based Door Lock System

PIC16F877A · AS608 Sensor · C

- Engineered a biometric access-control system using PIC16F877A and AS608 fingerprint sensor, enabling secure keyless entry with enrollment, verification, and error-handling firmware written in C.
- Integrated 16×2 LCD for real-time status display and automated a solenoid locking mechanism triggered on successful authentication, improving physical security.

Automatic Street Light System

LDR · Transistor Switching · Proteus

- Designed an LDR-driven ambient-light controller that autonomously toggles street lighting, eliminating manual switching and reducing energy waste.
- Simulated and validated full circuit behavior in Proteus prior to hardware deployment, achieving consistent performance across variable environmental conditions.

Inventory Management System

C++ · OOP · Console Application

- Built a console-based inventory application in C++ using OOP principles; implemented full CRUD operations, file-based data persistence, and fast-lookup reporting.

GCD Calculator — VHDL FSM

VHDL · Xilinx Spartan-3E FPGA

- Implemented a Greatest Common Divisor calculator as a two-process FSM in VHDL, synthesized and deployed on a Xilinx Spartan-3E FPGA using Xilinx ISE.

ADDITIONAL INFORMATION

Languages: Urdu (Native), English (Professional Working Proficiency) **Interests:** Embedded Systems Development, IoT Applications, Hardware Prototyping, Open-Source Electronics